

MCS-ResNet: A Generative Robot Grasping Network Based on RGB-D Fusion

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Abstract—RGB and Depth (RGB-D) modalities are widely used in planar grasp detection methods. However, most previous grasp detection approaches fail to fully exploit the distinct information contained in these two modalities and simply regard depth information as part of RGB information. To better integrate RGB-D information and improve the accuracy of planar grasp detection, a novel generative multimodal fusion with channel and spatial attention residual network (MCS-ResNet) is developed, which is specifically designed for 2-D planar grasping. The proposed network has two newly designed modules and a distinctive input method that expands depth information into three channels. First, we propose a novel RGB-D fusion module to fully extract information from both RGB-D modalities by disregarding secondary information and enhancing channel and spatial sensitivity. Then, we convert single-channel depth images into three-channel images to enable the network to capture richer depth features. Finally, a novel residual module is designed to make the model pay more attention to channel-wise information. Besides, to equip the model with classification capabilities, a multiobject solid waste grasp dataset with class information is created. In the comparison experiments, MCS-ResNet obtains 95.9% grasping accuracy and 97.8% classification accuracy (CA) on the solid waste grasping dataset with classification information (SWC) dataset and 98.5% grasping accuracy on the Cornell dataset, which reflects the superiority of our model. Finally, the real-world robotic experiments were conducted in a conveyor belt scenario using an edge computing device, reaching a grasping success rate of 95.3% and a classification success rate of 96.0%.

Index Terms—Generative network, grasp, grasp detection, RGB and Depth (RGB-D) dataset, RGB-D fusion, robot.

I. INTRODUCTION

ROBOTIC grasp has been widely utilized in industrial production, however, it is still a challenge for robots to achieve efficient and autonomous grasping. To accomplish grasping tasks in dynamically changing real-world scenarios, grasp detection methods must possess the capability to detect unknown objects and demonstrate robustness against environmental changes and noise interference. The primary challenge

lies in establishing the relationship between image and grasp poses, learning data features, and transferring them to novel objects.

Traditional methods such as modeling the environment, analyzing the dynamics and mechanics of object geometry, and determining object poses for grasping tasks consume significant computational resources. Remodeling the scene frequently in response to environmental changes is not feasible.

To achieve fast and accurate grasping, grasp detection methods based on deep learning [1], [2], [3] have been proposed. Among these methods, the 5-D representation method based on grasp rectangles is widely used. Through learning a large number of samples, the network acquires the ability to generate the required elements of the grasp rectangle and finally obtains the optimal grasp rectangle. Methods based on deep learning rely on grasp datasets; however, existing grasp datasets like [4], [5], [6], and [7] do not contain class information, making it difficult to achieve object detection while performing grasp detection using these datasets [8], [9]. Besides, research [10] suggests that utilizing RGB and Depth (RGB-D) information provides advantages over using RGB or depth information alone for grasp detection tasks. However, many works [11], [12] using RGB-D information merely treat depth images as an additional component of RGB images, without recognizing the differences between the two modalities, thereby leading to insufficient integration of information. Therefore, the methods to facilitate the full integration of RGB-D information need to be developed.

To improve the accuracy of grasp detection, realize high-precision grasping and classification prediction, and provide a feasible solution for solid waste sorting, we design a set of real-time 2-D planar grasp detection algorithms for robotic systems with visual servoing.

To equip the network with classification capability and enhance the grasping accuracy, we create a solid waste grasping dataset with classification information (SWC dataset) and propose a novel generative grasp model multimodal fusion with channel and spatial attention residual network (MCS-ResNet). This work specifically applies to 2-D planar grasping. First, the objects in the SWC dataset are categorized into nine major classes and 13 subclasses based on their material and shape, considering both the perspectives of classification for recycling and grasping. Second, we develop an MCS module to fully integrate RGB-D information and improve the channel and spatial sensitivity of feature maps. Moreover, to further improve the flow of multimodal feature information, traditional single-channel depth images are transformed into three-channel depth images. Third,

Received 17 July 2024; revised 28 October 2024; accepted 19 November 2024. Date of publication 25 December 2024; date of current version 6 January 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62073129 and Grant U21A20490, in part by the Project of the National Key Research and Development Program of China under Grant 2021YFC1910402, in part by Hunan Provincial Natural Science Foundation of China under Grant 2022JJ10020, and in part by the Scientific Research Project of Hunan Education Department under Grant 21B0330. The Associate Editor coordinating the review process was Dr. Ferdinanda Ponci. (Corresponding author: Qiaokang Liang.)

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Digital Object Identifier 10.1109/TIM.2024.3522430

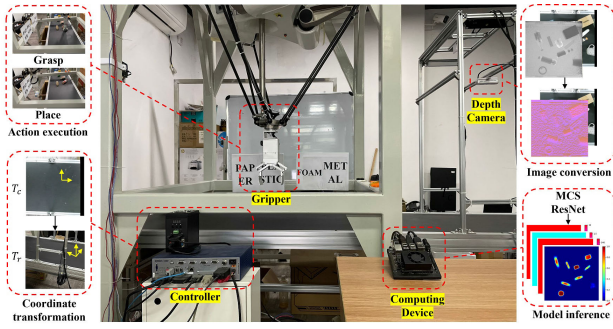


Fig. 1. Overview of the grasp detection system. The system first converts RGB-D images to RGB-HHA images acquired by the Depth camera and then sends them to MCS-ResNet deployed on the edge computing device for inference. Then, the controller converts the inference result from the image coordinates to the robot world coordinates. Finally, the gripper grasps the object and places it at the designated location according to the category.

we propose an effective residual module combined with the channel attention mechanism. The ablation experiments validate the effectiveness of the RGB-D fusion module, new input format, and novel residual module. Besides, comparative experiments on the SWC and Cornell datasets also show the superiority of our model. Finally, we conduct robotic grasping experiments by randomly placing solid waste on a conveyor belt. The overview of our grasp detection system is shown in Fig. 1.

The main contributions of this article can be summarized as follows.

- 1) We propose a novel fusion module that fully leverages information from RGB-D modalities, which enhances both channel and spatial sensitivity, thereby improving the accuracy of the grasp network. Besides, we convert the single-channel depth images into three-channel images at the input stage, enabling better integration of RGB-D information.
- 2) A multiobject solid waste grasping dataset with class information (the SWC grasp dataset) is created. This dataset consists of 2161 RGB-D images, and the objects are divided into nine major classes and 13 subclasses based on their material and shape.
- 3) We design a simple residual module to improve the attention of channel-wise information in the process of solving gradient vanishing and gradient explosion.
- 4) The proposed MCS-ResNet network achieves state-of-the-art performance on datasets and reaches high accuracy of grasping and classification in the real-time robot system.

The remaining sections of this article are organized as follows. Section II shows recent research work on grasp detection and RGB-D fusion methods. Section III gives the general problem formulation of planar grasp detection. Section IV presents methods we used, including the newly designed modules and MCS-ResNet structure. Section V describes the basic information about the SWC grasp dataset and the process of creating it. Section VI conducts the experiments, including experimental conditions, the establishment of the SWC grasp dataset, datasets experiments, and robotic experiments. Section VII provides a discussion of the number of modules

used as well as loss function weight assignments. Section VIII summarizes the text and describes the limitations of the methodology and directions for future work.

II. RELATED WORKS

A. Grasp Detection Methods

Grasp detection methods can be divided into two main categories: analytical and data-driven methods. The analytical method is a complex coding approach [13], which involves kinematic, dynamic, and mechanical analyses of object geometry and pose to ascertain the grasp location. Approaches like [14] are designed for specific robots, grasping environments and tasks. This method needs to satisfy the kinematics and dynamics formulas, and a large number of constraint conditions, which makes the calculation difficult and inefficient. At the same time, for unstructured environments characterized by a lack of clear structure and regularity, along with numerous uncertainties and complexities, more sophisticated analytical techniques are needed to develop solutions. This necessitates the incorporation of additional data and robot physical parameters, rendering the modeling process more challenging.

Data-driven method is mainly based on deep learning [15], [16], which is divided into two types: discriminative [1], [16] and generative [2], [12], [17] methods. The discriminative method directly classifies and discriminates grasping gestures based on existing data and knowledge, which usually includes several steps of feature extraction, feature representation, classification, and discrimination. For example, Chu et al. [18] included a grasp region proposal network and divided the grasping task into regression over the bounding box parameters and classification of the orientation angles. The generative method refers to learning the distribution probability of the data to generate new data based on the shape, size, center of gravity, and other characteristics of the object through mathematical modeling and optimization algorithms, thus simulating the generation process of real data. Redmon and Angelova [19] applied a single network to the data and predicted grasp rectangle directly. Kumra et al. [10] inferred multiple grasp coordinates from the output in one shot, which improves efficiency.

B. Attention Mechanisms

Nowadays, attention mechanisms [20], [21], [22] are widely used in computer vision tasks. The attentional mechanisms make the network pay more attention to more important information in the spatial or channel dimension and suppress the unimportant one by assigning feature weights. The famous [23] proposes a lightweight and efficient squeeze-and-excitation (SE) channel attention module that reallocates the weights of channel dimensions by simple average pooling operation and obtains significant results. Inspired by the idea of Hu et al. [23] and Park [24] proposed a bottleneck attention module (BAM), BAM enhances the network's attention to key features by concatenating channel attention and spatial attention. Liu et al. [25] designed self-calibrated convolutions consisting of a combination of multiple convolutional attentions, which overproduces a global feeling field without adding additional parameters and computational effort.

C. RGB-D Fusion Methods

RGB-D images are the most commonly used data types for grasping tasks, and relevant studies have proved that using RGB-D fusion image information has greater advantages in grasping inference than using RGB image or depth image information alone [10], [26]. Related multimodal fusion methods toward dense image prediction are mainly divided into aggregation-based fusion [27], [28], [29] and alignment-based fusion [30], [31], [32]. Aggregation-based fusion methods apply averaging [33] and concatenation [34] to fusion feature maps. For example, Xu et al. [35] proposed the information measurement for unsupervised learning by concatenating source images. The alignment-based fusion methods adopt regulation loss to align subnet embeddings while maintaining the full propagation for each of them. Wang et al. [31] proposed a possible solution to alleviate the problem of simply focusing on the whole distribution, which may overlook specific patterns within each modality. Fusion methods could also be specified as early, intermediate, and late fusion depending on the fusion stage [36], [37], [38]. Besides these two main categories, Wang et al. [39] proposed a channel exchange method, guaranteeing sufficient information flow. Xue and Marculescu [40] proposed dynamic multimodal fusion, which provides a flexible way to control how and when the other modality comes in.

III. PROBLEM FORMULATION

Redmon and Angelova [19] and Kumra and Kanan [41] have proposed a famous 2-D image grasping representation, which is represented by $G = (P, \Theta, W, Q)$. In this work, we adopt an improved 2-D grasp representation method that includes classification information to achieve pixel-wise grasping estimation, which can be described as

$$\mathbf{G}_r = (\mathbf{P}, \Theta_r, \mathbf{W}_r, \mathbf{Q}, \mathbf{C}) \quad (1)$$

where $P = (X, Y, Z)$ represents the center position of the gripper, Θ_r is the gripper orientation angle around the Z -axis, $\mathbf{W}_r$ is the required opening width for the gripper, Q is the grasp quality score, and C is the category of the object corresponding to the grasp.

In an n -channel image $\mathbf{I} \in \mathbb{R}^{n \times h \times w}$ with height h and width w , a grasp can be defined as

$$\mathbf{G}_i = (x, y, \theta_i, \mathbf{w}_i, \mathbf{q}, \mathbf{c}) \quad (2)$$

where (x, y) represents the grasp center, θ_i is the grasp angle at the i_{th} pixel with the range of $[-(\pi/2), (\pi/2)]$, w_i is the needed grasp width with the range of $[0, W_{max}]$ in image coordinates, and q is the quality score of this grasp, with values ranging between 0 and 1, the higher the score indicates the higher success rate of the grasp. The local maximum algorithm will obtain the pixel coordinates with the highest grasp quality score in the local region. c represents the classification information for grasping objects, with each numerical value corresponding to a specific category of object.

To apply the grasp obtained in the image space, the following transformations to convert the image coordinates to the robot's frame of reference are used:

$$\mathbf{G}_r = \mathbf{T}_{rc}(\mathbf{T}_{ci}(\mathbf{G}_i)) \quad (3)$$

where $\mathbf{T}_{ci}$ is the transformation that converts 2-D image space to 3-D camera space by using camera intrinsic parameters and $\mathbf{T}_{rc}$ is the transformation that converts 3-D camera space to robot space with the camera calibration results.

IV. METHOD

A. MCS Module: RGB-D Fusion Module Based on Channel Exchange

Recent works [11], [42] concatenate a three-channel RGB image with a single-channel depth image as a four-channel image and then input into the network, which overlooks important edge information in the depth image and cannot capture the differentiation information across modalities. To fully use the information from different modalities, we designed an RGB-D fusion module, which combined channel attention, spatial attention, and channel exchange together. The architecture is shown in Fig. 2. The RGB-D feature with the same shape $C \times H \times W$ is input into the module and the channel attention extracts channel-wise weights, which are used to compare with a specified threshold in the mm module, as shown in Fig. 3. After performing channel exchange operations in the mm module, the unimportant channels are discarded and then we apply spatial attention to enhance the spatial sensitivity of features. Through these operations, we have enhanced the original features both in channel-wise and spatial-wise dimensions and generated new feature maps with the same shape.

The channel attention is inspired by Wang et al. [43], which consisted of a convolution layer with a 3×3 convolution kernel, a Batch normal function, and a global average pooling layer, followed by a sigmoid function. The spatial attention first captures the mean and maximum values of the input feature map along the spatial dimensions and then concatenates these values before the convolution layer with the 3×3 convolution kernel, after the convolution layer, we apply the Batch normal function and a sigmoid function. The channel attention and spatial attention outputs can be described as

$$\begin{cases} \mathbf{F}_{out}^c = \text{Sigmoid}(f_{avgpool}(f_{bn}(f_{conv}(\mathbf{F}_{in}^c)))) \\ \mathbf{F}_{cat} = \text{Concat}(f_{mean}(\mathbf{F}_{in}^s), f_{max}(\mathbf{F}_{in}^s)) \\ \mathbf{F}_{out}^s = \text{Sigmoid}(f_{bn}(f_{conv}(\mathbf{F}_{cat}))) \end{cases} \quad (4)$$

where $\mathbf{F}_{out}^c, \mathbf{F}_{in}^c, \mathbf{F}_{out}^s$, and $\mathbf{F}_{in}^s$ are channel and spatial attention's output and input feature maps, respectively. Sigmoid, f_{avg} , f_{bn} , f_{conv} , f_{mean} , and f_{max} are the sigmoid function, global average pooling, Batch normal function, convolution, mean, and max operations.

In the mm module, the RGB feature and Depth feature map perform channel exchange operations based on a set of conditions: if the weight of a feature map at the c th channel is below a certain threshold for one feature map, while the weight of the other feature map exceeds another threshold, then the c th channel of the former is replaced by that of the latter. For the c th channel, the condition

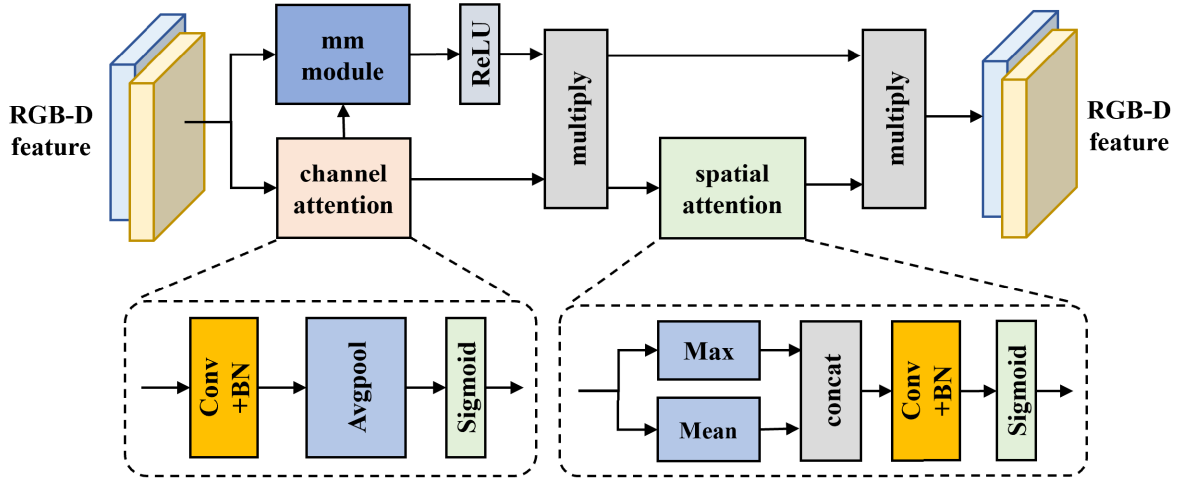


Fig. 2. Structure of the proposed MCS module. The MCS module mainly consists of an mm module, a channel attention module, and a spatial attention module. The MCS module performs deep feature extraction and fusion of RGB-D features, making fuller use of the information contained in these two different modalities. The output shape of the MCS module remains consistent with the input shape, allowing it to function as a plug-and-play component.

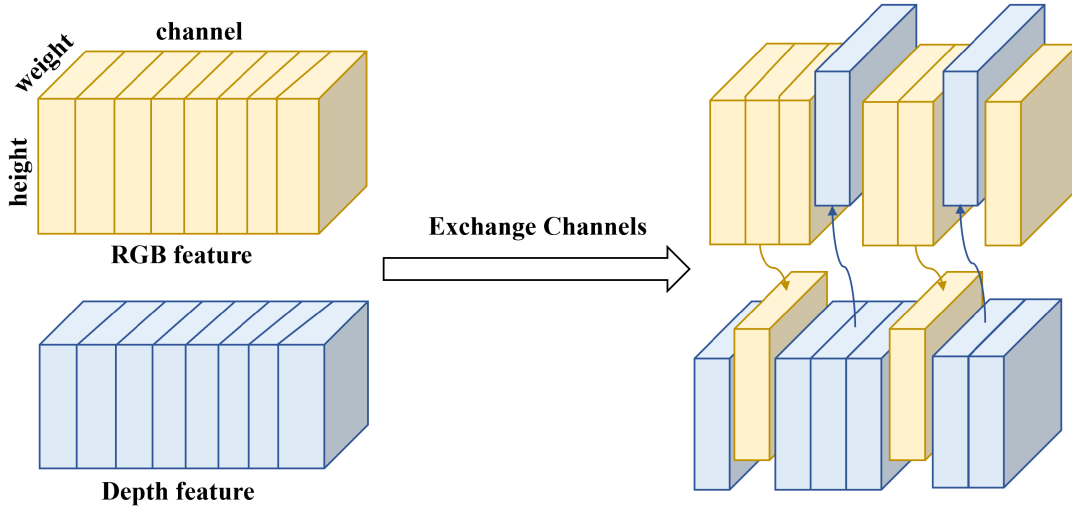


Fig. 3. Proposed mm module. The mm module is a crucial component of the MCS module, where RGB-D features exchange channels based on the weight information of channel dimension. Unimportant information is discarded in this process.

can be described as

$$w_1 < \text{threshold}_1 \ \& \ w_2 > \text{threshold}_2 \quad (5)$$

where w_1, w_2 are weights extracted by channel attention, $\text{threshold}_1, \text{threshold}_2$ are specified values, and the subscripts represent different modalities. In this work, we set the threshold by

$$\text{threshold} = r \times f_{\text{sort}}(\mathbf{W}) \quad (6)$$

where f_{sort} is a function for sorting in ascending order and $\mathbf{W}$ is obtained weight information. After this operation, the threshold is set to the r th percentile position of the sorted sequence of $\mathbf{W}$ in ascending order and the r is set to 0.35.

Overall, the whole process can be written as

$$\mathbf{F}_{\text{out}} = f_s(\text{ReLU}(f_{ce}(f_c(\mathbf{F}_{\text{in}})))) \quad (7)$$

where $\mathbf{F}_{\text{out}}$ and $\mathbf{F}_{\text{in}}$ are output and input feature maps and f_s, f_{ce} , and f_c are spatial attention, channel exchange, and

channel attention operations, respectively. ReLU is an activation function.

In the MCS module, RGB-D feature maps with the same shape are first acquired by channel attention to obtain the weights of channel dimensions, and then the weight information is used to perform channel exchange operations in the mm module. The exchanged feature maps are then fed into the spatial attention module to enhance the spatial sensitivity of the feature maps. The final output is an RGB-D feature map that maintains the same size as the inputs.

B. Sca-Res: Efficient Residual Module

He et al. [44] indicated that residual layers can effectively alleviate the issues of vanishing gradients and exploding gradients. Each residual block typically consists of two primary branches: the main branch and the residual branch, which can be described as

$$H(x) = F(x) + x \quad (8)$$

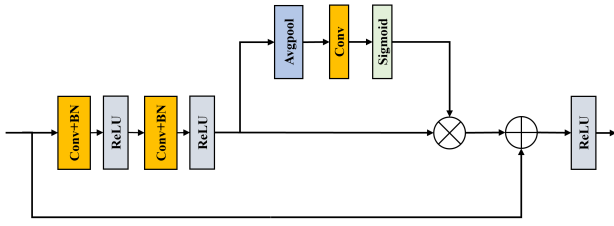


Fig. 4. Sca-res module: an efficient residual module.

where $H(x)$, $F(x)$, and x are the output, the residual, and the input of a residual block, respectively.

In this article, we design an efficient residual module, as shown in Fig. 4. First, two convolution layers with 3×3 convolution kernel are used to extract basic features. Batch normal and ReLU functions are used after each convolution layer. Then, we use a global average pooling layer followed by a convolution layer before a sigmoid function to enhance the channel sensitivity. The residual is obtained by multiplying the basic features with channel weights. Finally, a ReLU function is employed to perform element-wise addition of the input feature map and the residual.

In proposed residual module, $F(x)$ can be written as

$$F(x) = \text{Sigmoid}(f_{\text{conv}}(f_{\text{avgpool}}(F))) \otimes F \quad (9)$$

where F is given by

$$\begin{cases} F = \text{ReLU}(f_{\text{bn}}(f_{\text{conv}}(F'))) \\ F' = \text{ReLU}(f_{\text{bn}}(f_{\text{conv}}(X))) \end{cases} \quad (10)$$

where $\otimes$ means element-wise multiplication and X is the input of the residual block.

By applying the channel attention mechanism on the residual term, the network can be made to pay more attention to the important features, thus improving the performance of the model.

C. HHA Encoding Depth Images: A Three-Channel Depth Images Format

Given the requirement of the MCS module for input feature maps to possess uniform shapes, we transform single-channel depth images into three-channel horizontal disparity, height above ground, and angle with gravity (HHA) images, which is proposed by Gupta et al. [45]. To achieve this transformation, the camera's intrinsic parameters are required. The HHA encoding format actually spreads the information from a single-channel depth image into three channels, aligning with the channel structure of RGB images. This structural symmetry facilitates more comprehensive cross-modal information fusion. The HHA encoding format enhances the network's capability to extract depth feature information, albeit at the expense of increased computational resources.

D. Network Architecture

The overall network architecture of MCS-ResNet is shown in Fig. 5, which improved the GR-Convnet [10]. At the input stage, a couple of RGB images and an HHA Depth image with a size of 224×224 are fed into the network separately rather

than concatenated together. Then, three repeating modules are used, in which the MCS module is applied following a convolution layer and a Batch normal function.

To simplify the network architecture, the results of the last MCS module are concatenated along the channel dimension. The size of the output is $256 \times 56 \times 56$. Then five Sca-res blocks are employed to alleviate gradient vanishing and in the Sca-res blocks, and the size of the tensor is maintained as $256 \times 56 \times 56$. Finally, three transposed convolution layers are added to turn the tensor size back to $32 \times 224 \times 224$.

The outputs of the network consist of five components: the opening width of the gripper, $\cos 2\theta$, $\sin 2\theta$, the quality of grasp, and the category of the grasp object. Each of them is generated by a convolution layer with the kernel size of 1×1 . The grasp angle can be calculated by $\theta = 1/2 \arctan(\sin 2\theta / \cos 2\theta)$ and the center position of the gripper, denoted as P , is obtained in the heatmap of Q . We use Gaussian filtering to postprocess the output grasp quality Q , then get the angle, width, and class information of the pixel with the largest local Q value.

E. Loss Function

We apply Smooth $L1$ loss as a loss function, which is more stable and robust in the training process. The total loss function can be described as

$$\text{Loss} = l_q + l_w + l_{\sin} + l_{\cos} + l_c \quad (11)$$

with l

$$l(\hat{G}_i, G_i) = \begin{cases} 0.5(\hat{G}_i - G_i)^2, & \text{if } |\hat{G}_i - G_i| < 1 \\ |\hat{G}_i - G_i| - 0.5, & \text{otherwise} \end{cases} \quad (12)$$

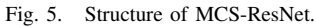
where l_q , l_w , $l_{\sin}$, $l_{\cos}$, and l_c is the Smooth $L1$ loss of the quality of grasp, the opening width of the gripper, $\sin 2\theta$, $\cos 2\theta$, and the category of the grasp object, respectively. $\hat{G}_i$ is the predicted value and G_i is the ground-truth value.

The ground truth of the opening width of the gripper $\sin 2\theta$, $\cos 2\theta$ is obtained from the coordinates of the labeled grasp rectangle, and within the labeled grasp rectangle, the ground-truth value of Q is set to 1.

V. SWC GRASP DATASET

For the model to achieve higher grasp and classification accuracy (CA) in the visual servo system with conveyor belts as the background, we create a new multiobject solid waste grasping dataset with the conveyor belt as the background, and its grasp rectangle annotations are consistent with the Cornell dataset [4], which is named the SWC grasp dataset. The dataset includes common household solid waste objects such as plastic bottles and paper boxes, and the detailed information of the dataset is shown in Fig. 6. Objects in the SWC grasp dataset are classified into two levels. The level 1 categorizes the objects into nine classes based on their material. The level 2 further subdivides these classes into 13 categories according to their shape. The proportions of various level 1 objects are as shown in Fig. 7.

We develop an auxiliary tool with directed rectangles to assist in annotating the original RGB images. The annotation interface is shown in Fig. 8. In addition to annotating



All RGB-D images were registered using the method proposed by Kim et al. [46]. The correspondence between the RGB-D images can be established using a transformation matrix. The transformation matrix to convert the RGB image coordinates (x, y) to depth image coordinates (X, Y) is given as follows:

We use the checkerboard to find the pairs of corresponding points between the RGB-D images. Using these points, we solve for the unknown parameters of the transformation matrix T . Once the transformation matrix T is determined, we can align the RGB images to the depth images.

A. Datasets and Experimental Setting

2) *Hardware Platform:* We train the networks on an NVIDIA GeForce RTX 4090 GPU and a 13th Gen Intel¹ Core² i9-13900K CPU. Additionally, in robotics experiments,

²Trademarked.

Hyperparameter	Value
Batch Size	8
Num Workers	8
Epochs	50
Learning Rate	0.001
Optimizer	Adam

The experimental robot platform is the Omron Hornet 565. The Hornet 565 is a parallel robot designed for high-speed picking and packaging applications. The overall solid waste sorting platform is a robot system with visual servoing.

- 1) The angle difference between the predict grasp rectangle and the ground truth is less than 30° .
- 2) The intersection over union (IoU) score between the predict grasp rectangle G_i and the ground truth $\hat{G}_i$ is larger than 0.25

B. Experiments on the Cornell Dataset

The hyperparameters used during training are listed in Table I, consistent with the comparative methods for a fair performance comparison. We evaluated the performance

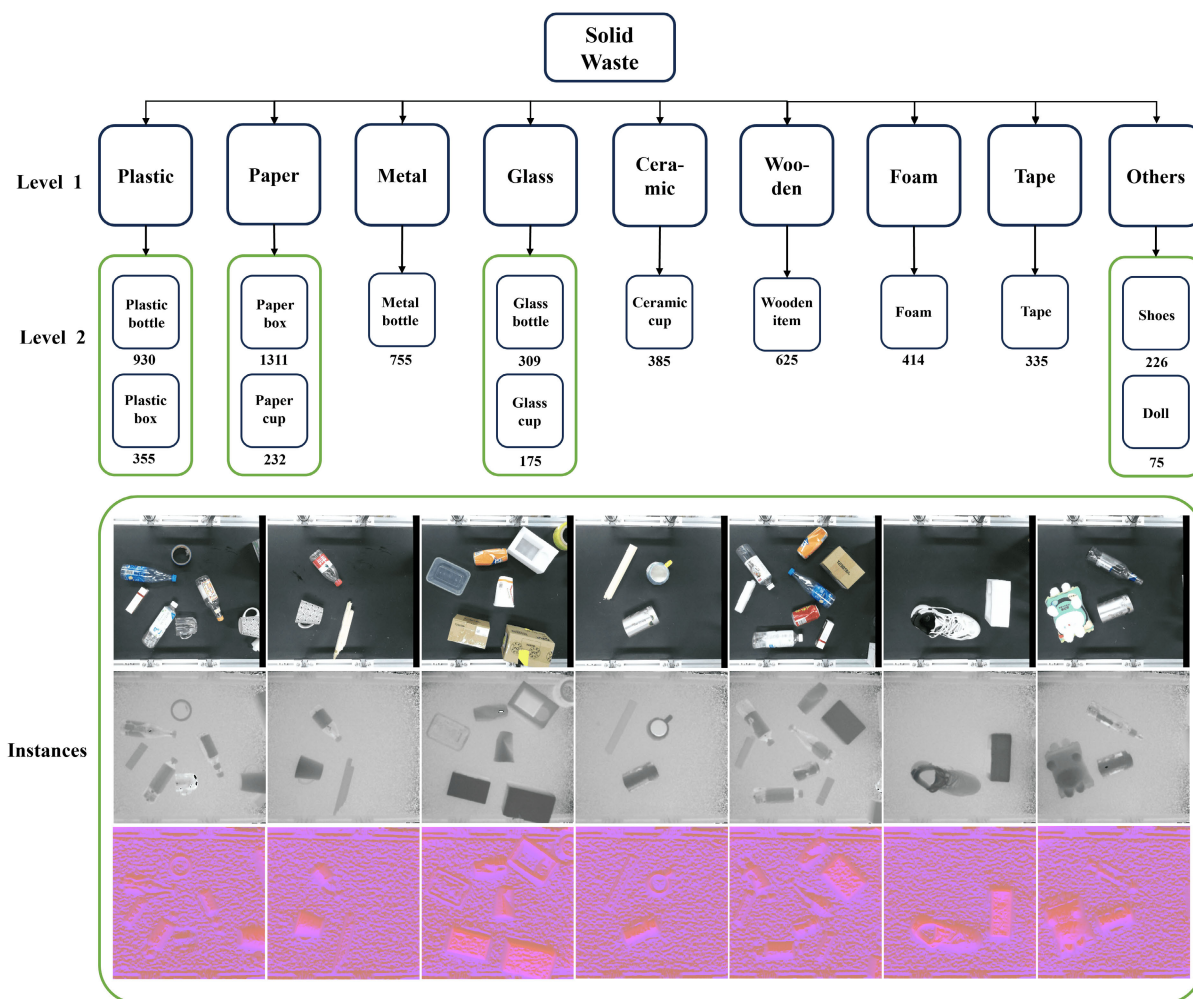


Fig. 6. Detail information of our dataset. Solid waste is categorized into nine major classes and 13 subclasses, and each subclass is marked with a specific quantity in the figure. Instances give some RGB, HHA, and visualizing Depth images in the SWC dataset.

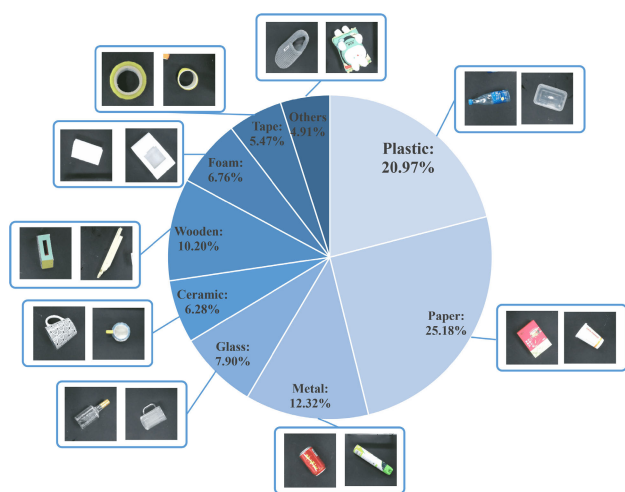


Fig. 7. Distribution and examples of categories in the SWC dataset.

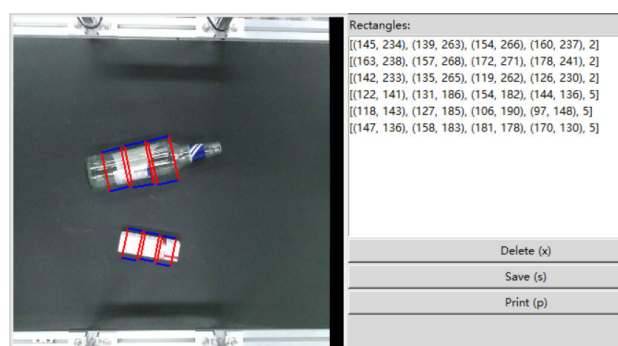


Fig. 8. Annotation interface with directed rectangles. It allows for simultaneous annotation of grasp rectangles and the corresponding object categories.

of MCS-ResNet on the Cornell dataset and compared it with recently proposed networks. Given that the Cornell dataset does not contain classification information, the classification

head in the network and the classification component in the loss function are disregarded.

In the experiment, data augmentation is implemented by using random cropping, scaling, and rotation, while only positive labels in the dataset are considered. We conducted cross-validation setup by image-wise (IW) and object-wise (OW) to divide the Cornell dataset and the detection accuracy is

TABLE II
COMPARISON RESULTS ON THE CORNELL DATASET

Methods	Params	IW(%)	OW(%)
MultiGrasp [19]	7.4M	88.0	87.1
GGCNN [47]	3.7K	73.9	69.0
GR-ConvNet [10]	1.9M	97.7	96.6
SE-ResUNet [11]	24.0M	98.2	97.1
SKGNET [12]	33.4M	99.1	98.4
MCS-ResNet	7.0M	98.5	97.5

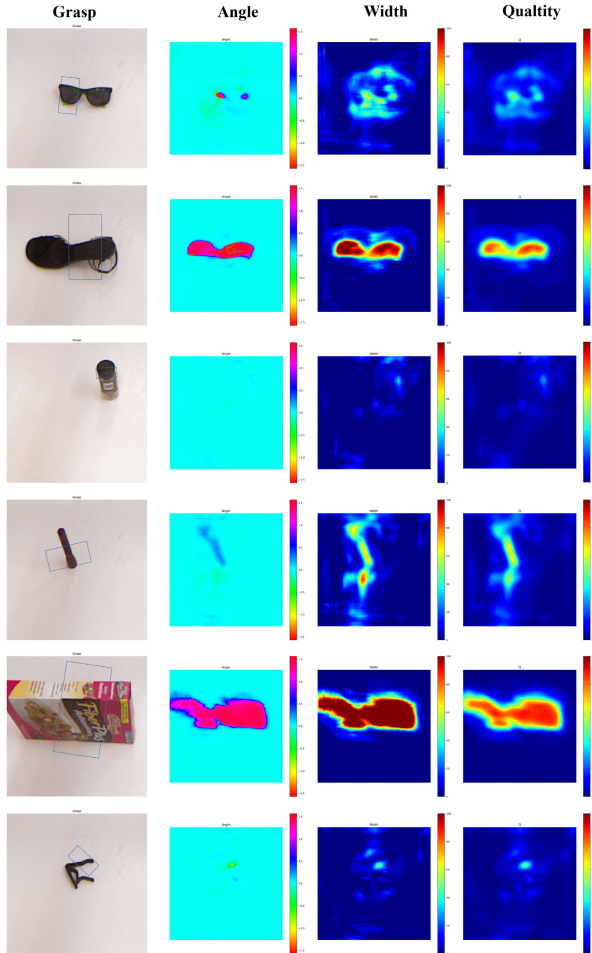


Fig. 9. Detection results on the Cornell dataset, including visualized grasp rectangles, grasp quality heatmap, grasp angle heatmap, and grasp width heatmap.

presented in Table II. The visualization results of MCS-ResNet on the Cornell dataset are shown in Fig. 9.

It is evident that compared with other methods, the MCS-ResNet exhibits superiority in accuracy due to the effective fusion of color and depth image information, as well as the adoption of more efficient residual blocks. Additionally, our model has a total of 7 003 266 parameters. Although the SKGNET [12] demonstrates higher precision in both IW and OW accuracy, its parameter count is nearly five times greater than ours, making deployment on edge computing devices significantly more challenging.

C. Experiments on the SWC Dataset

We conduct experiments on the SWC dataset with an existing generative grasp network. The training and testing sets

TABLE III
NETWORK HYPERPARAMETERS TRAINED ON THE SWC DATASET

Hyperparameter	Value
Batch Size	16
Num Workers	8
Epochs	200
Learning Rate	0.001
Optimizer	Adam

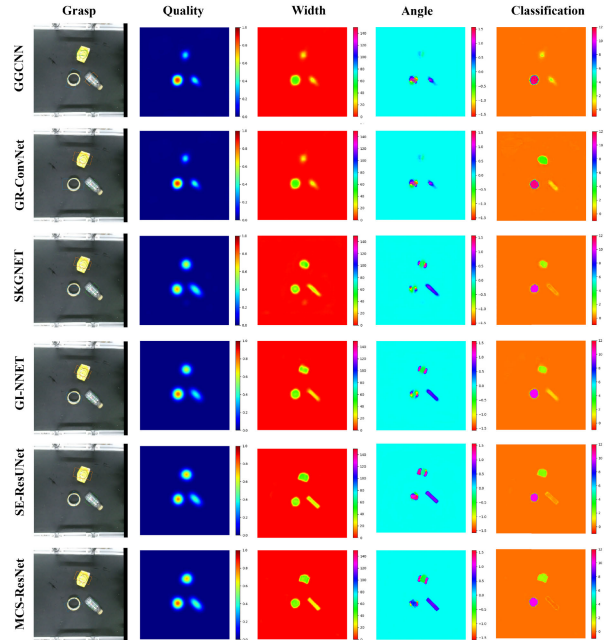


Fig. 10. Comparison studies on the SWC dataset, including visualized grasp rectangles, grasp quality heatmap, grasp angle heatmap, grasp width heatmap, and grasp classification heatmap.

are divided according to 8:2. Considering that the SWC dataset is a larger-scale multiobject grasping dataset, we increase the batch size and the number of epochs to enhance the training efficiency and performance of the model. The hyperparameters used during training are listed in Table III.

Considering that other grasp detection networks lack classification functionality, we add a classification head at the output stage similar to our model. Table IV shows input size, grasp accuracy (GA), CA, parameter quantity, and inference time of these networks on the SWC dataset. The HHA Depth images are exclusively adopted in our model, as other networks do not meet the input requirements.

The visualization results are displayed in Fig. 10, with the first row showing the best candidate grasp rectangle, followed by the grasping quality, angle, width, and classification, respectively. These results clearly demonstrate the superiority of our model in grasp detection and classification.

D. Ablation Experiments

The ablation experiments are performed to validate the effectiveness of the proposed modules. In summary, we conducted several experiments to validate the effectiveness of the MCS module, the Sca-res module, and the HHA format input, respectively.

TABLE IV
PERFORMANCE COMPARISON OF MODELS ON THE SWC DATASET

Methods	Input Size	GA(%)	CA(%)	Params	T1(ms) ¹	T2(ms) ²
GGCNN [47]	224 × 224	69.3	48.1	3.7k	13.09	4.35
GR-ConvNet [10]	224 × 224	94.0	93.0	1.9M	42.22	31.08
SKGNET [12]	288 × 288	90.1	86.2	33.4M	253.26	188.28
GI-NNet [48]	224 × 224	91.9	80.4	59.4K	36.54	26.71
SE-ResUNet [11]	224 × 224	94.1	97.5	24.0M	51.83	68.54
Ours	224 × 224	95.9	97.8	7.0M	48.72	54.19

¹ T1 denotes the inference time on the NVIDIA 4090.

² T2 denotes the inference time on the edge computing device Jetson Xaiver NX.

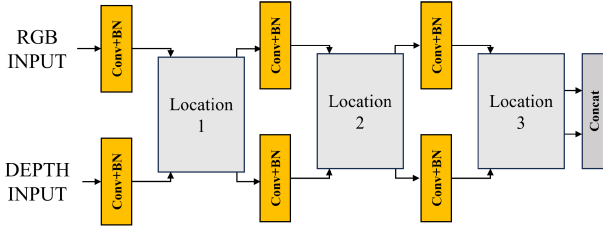


Fig. 11. Ablation experiments on the location and quantity of the MCS module. Locations 1, 2, and 3 are the locations where the MCS module can be applied.

TABLE V
EXPERIMENTAL RESULTS FOR DIFFERENT LOCATIONS
AND NUMBERS OF MCS MODULE USAGE

Location Combinations	GA(%)	CA(%)
1	93.5	94.0
2	91.1	94.8
3	93.9	97.4
1+2	94.4	97.7
1+3	93.9	96.3
2+3	94.3	96.4
1+2+3	95.9	97.8

We first conduct ablation experiments on the position and quantity of the MCS module. As illustrated in Fig. 11, we experiment after each convolution and the BN layer in the network encoder. For locations where the MCS module is not used, we add the RULE function for activation after the BN layer. We conduct this part of the ablation experiment using the controlled variable method, and the Sca-res module and the HHA format are employed. The final results are shown in Table V. The results indicate that applying the MCS module structure at all three potential locations yields the highest GA and CA, which is consistent with our initial hypothesis.

Besides, we also embed the MCS module into different grasp detection networks, specifically, we apply the MCS module to the encoding stage of these networks. The experimental results are shown in Table VI. The implementation details vary from model to model, resulting in varying increases in accuracy after using the MCS module. However, overall both the GA and the CA consistently improve. This proves the versatility of the MCS module for 2-D planar grasp detection tasks.

In addition, experiments for the Sca-res module are carried out. We replace the Sca-res module in MCS-ResNet with some widely used residual modules [11], [42], [44], [49] and compare the GA and CA on the SWC dataset. The results are

TABLE VI
MODELS' PERFORMANCE IMPROVEMENT AFTER
EMBEDDING THE MCS MODULE

Methods	GA(%)	CA(%)
GGCNN [47]	72.7 (+3.4)	51.1 (+3.0)
GR-ConvNet [10]	95.2 (+1.3)	96.9 (+3.9)
SKGNET [12]	92.3 (+2.1)	96.6 (+10.4)
GI-NNet [48]	92.0 (+0.1)	87.8 (+7.4)
SE-ResUNet [11]	95.1 (+1.0)	98.1 (+0.6)

TABLE VII
COMPARISON RESULTS OF APPLYING DIFFERENT
RESIDUAL MODULES IN MCS-RESNET

Residual module	GA(%)	CA(%)
GR-ConvNet V2 [42]	92.1	94.1
SE-ResUNet [11]	93.1	95.6
ResNet18 [44]	95.6	97.0
ResNet50 [44]	92.1	95.0
ResNet V2 [49]	94.0	95.3
Sca-res(Ours)	95.9	97.8

TABLE VIII
RESULTS OF THE ABLATION EXPERIMENTS ON THE PROPOSED METHODS

Network	GA(%)	CA(%)
Baseline	93.1	94.3
Baseline+HHA Format	94.8	95.5
Baseline+MCS-Module+Sca-res	93.7	96.6
Baseline+MCS-Module+HHA Format	95.6	97.0
Baseline+Sca-res+HHA Format	95.7	95.4
MCS-ResNet	95.9	97.8

shown in Table VII and the results show the superiority of our proposed Sca-res module compared to common residual modules in the field of 2-D planar grasp detection.

Finally, we conduct ablation experiments to evaluate the proposed methods in MCS-ResNet. By removing or replacing these modules one by one, we can evaluate the impact of each module on the overall performance and thus further determine the contribution of each module. Specifically, we compare the GA and CA after removing the MCS module from the existing network architecture. In addition, the Sca-res module is replaced with the classical residual block used in [10]. Similar to other research work, depth images are inputted to the model as single-channel representations instead of HHA format, meanwhile modifying the parameters of the first convolution layer to accommodate the input. The ablation experiment results are shown in Table VIII.

It is clear that the methods proposed in this article effectively improve GA and CA, albeit with a slight increase in the computational and parameter requirement.

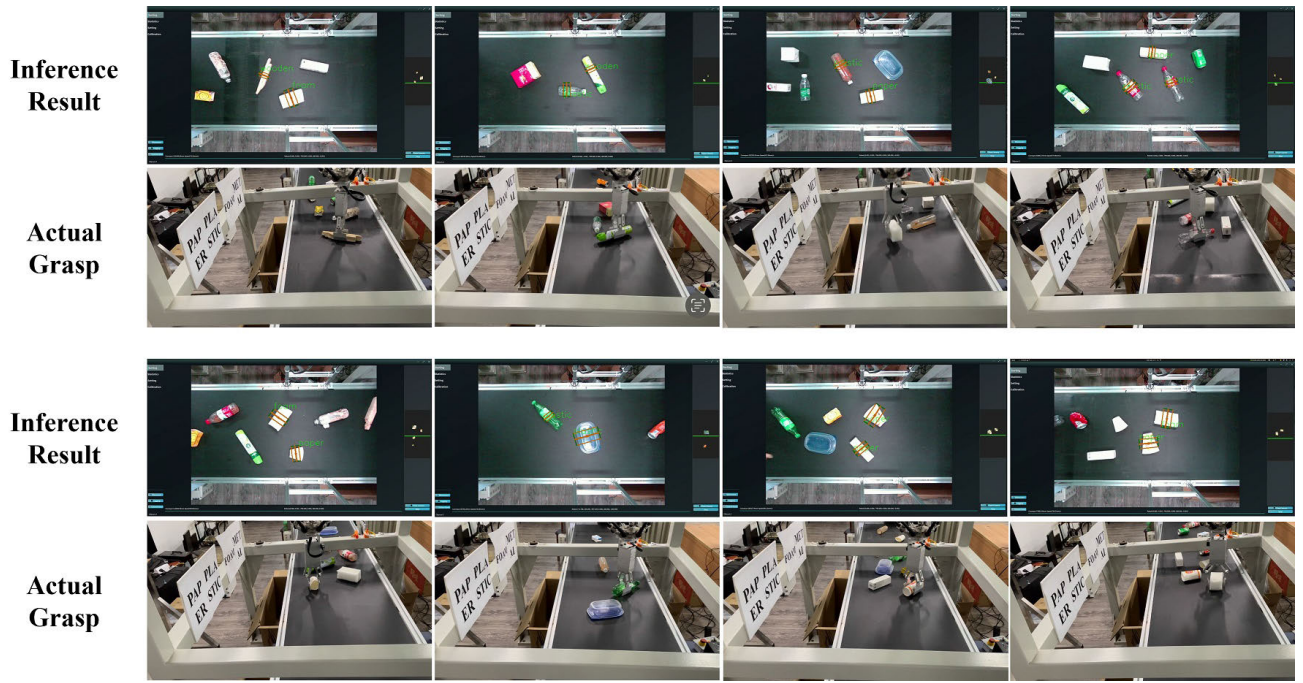


Fig. 12. Robotic grasping in a real-world environment. The inference results refer to the grasp rectangle with category information generated by the model. Actual grasp refers to the action of the gripper grasping the object. The vertically aligned inference result corresponds to the actual grasp.

E. Robotic Grasping

To evaluate the performance of our model in the real world, we conducted robotic grasping experiments against a conveyor belt background. Specifically, an Intel RealSense D455 camera and a parallel gripper are equipped on a solid waste sorting platform. In each actual grasp, MCS-ResNet first generates the optimal grasp rectangle based on the images obtained from the camera. Subsequently, the image coordinates are then transformed into the world coordinate system of the conveyor belt. Finally, the gripper executes the grasping action, and the objects are placed at specified positions according to their categories.

To explore the possibility of deploying the model on embedded platforms, we optimize our model with TensorRT and conduct experiments on Jetson Xavier NX. Based on the acceleration of TensorRT, the model achieves an average inference speed of 62 ms per image, which suggests that MCS-ResNet meets the requirement for real-time execution.

We conducted a total of 15 robotic experiments on the conveyor belt, each time placing ten objects of different shapes and categories at random angles. The experimental scene is shown in Fig. 12. The grasping success rate and CA for each robotic experiment are shown in Table IX.

Failure cases can be categorized into two types: grasping failures and classification errors. Classification errors result solely from incorrect model inference. Grasping failures can generally be attributed to two main reasons: software errors caused by incorrect model inference and hardware errors inherent to the robotic system. The failure cases are shown in Fig. 13.

By randomly placing solid waste on the conveyor belt, the system achieved a grasping success rate of 95.3% (143/150) and a classification success rate of 96.0% (144/150). Overall,

TABLE IX
GRASPING SUCCESS RATE AND CA IN ROBOTIC EXPERIMENTS

Experiment Number	Successful Grasps	Correct Classifications
1	8	9
2	10	10
3	9	9
4	10	10
5	9	10
6	10	9
7	10	9
8	9	10
9	10	10
10	10	9
11	10	10
12	9	10
13	10	9
14	10	10
15	9	10

our model performs well in real-world tasks of solid waste grasping and classification.

VII. DISCUSSION

In this section, we discuss the number of MCS modules used, the proportional distribution of the loss function, and the choice of the origin network.

The results in Table V show that applying an MCS module after each convolution layer in the encoding stage works best, and we discuss the number of MCS modules while maintaining this correspondence. The experiment results show a decrease in both GA and CA with two MCS modules (92.5% and 95.0%) and four MCS modules (94.1% and 94.9%). The reason may be that when using two MCS modules, the encoding process does not fully compress and extract the key features of the data; when using four MCS modules, too many channel exchange operations will discard the channels that

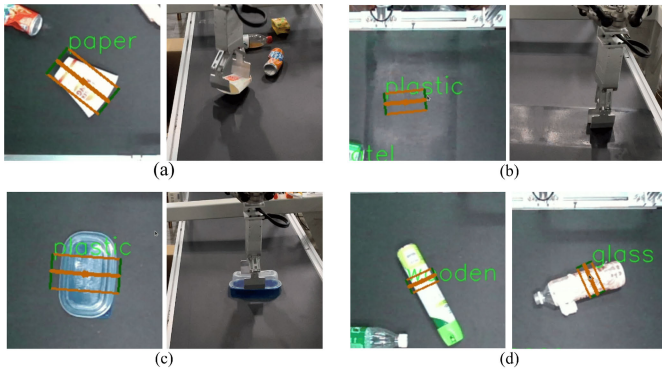


Fig. 13. Failure cases in robotic experiments. (a) Grasp failure due to incorrect angle generated by the model. (b) Grasping failure due to conveyor belt reflection. (c) Grasping failure due to gripper positioning error. (d) Classification errors.

themselves contain key information, ultimately leading to poor detection accuracy.

The loss function consists of Smooth $L1$ loss of the quality of grasp, the opening width of the gripper, $\sin 2\theta$, $\cos 2\theta$, and the category of the grasp object. Consistent with most related work, these five components are kept equally weighting ratios. Yu et al. [12] assigned a higher weight to the quality of grasp in the loss function (1.2). We conduct experiments using MCS-ResNet with the same parameter settings, while the accuracy decreases slightly (GA: 95.0%; CA: 96.2%). The reason for the insignificant change is that the width and angle of the grasp together determine the grasp rectangle (the length of the gripper is generally fixed) and the grasp width and angle are generated at the pixel points with the highest localized grasp quality. They are not in a parallel relationship and are hard to measure with different weights.

The results in Table IV demonstrate that SE-ResUNet [11] outperforms GR-ConvNet [10]. However, we do not select SE-ResUNet as the origin network. We apply the proposed MCS module, Sca-Res, and HHA encoding format to SE-ResUNet for experimentation, achieving a GA of 94.5% and a CA of 97.5%. The model using SE-ResUNet as the origin network shows performance second only to our proposed MCS-ResNet. Nevertheless, the parameter count of SE-ResUNet is 12 times that of GR-ConvNet, and the model with SE-ResUNet as the origin network contains 97 250 503 parameters, which is 13 times that of MCS-ResNet. The excessive number of parameters hinders the deployment of the model on edge computing devices with limited computational power.

VIII. CONCLUSION AND FUTURE WORK

In this article, we propose a generative grasp detection network, MCS-ResNet. To fully utilize RGB-D information, a novel fusion module and different input formats that convert RGB-D to RGB-HHA are employed. Besides, to enhance the ability of information flow, we design a novel residual module. Additionally, we create a multiobject solid waste grasping dataset with class information (SWC dataset) to empower the network with classification capability. Then, we evaluate the proposed MCS-ResNet on the SWC and Cornell datasets and

the results demonstrate its advantages over existing networks in the GA and CA. Ultimately, the robotic grasping experiments prove its effectiveness and the inference speed meets real-time requirements. In general, the proposed network effectively integrates information from different modalities and performs well in grasp detection tasks. It also provides a potentially feasible solution for the automated sorting of solid waste.

However, there are still limitations to our work. First, the model's classification of objects is limited to 13 categories. To recognize new categories, it needs to be retrained or learned using incremental learning methods; second, the construction of the SWC grasp dataset is time-consuming and labor-intensive, which makes it difficult to create a large-scale real-world grasp dataset. Third, the methods we propose are primarily designed to improve the accuracy of the grasp network, with less emphasis on lightweight and high-speed inference of the model. Future work will focus on three main aspects: developing efficient semi-supervised learning methods, building robotic systems that can continuously learn new knowledge using incremental learning, and developing lightweight grasp detection networks. Through semi-supervised learning and incremental learning, models can continuously learn new knowledge at a low cost and adapt to different complex scenarios, while models with smaller parameter counts are favorable to be deployed on computing devices with more constrained computational resources.

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